EWB CLOUD LABS • STUDENT PROJECT WORKBOOK

AWS VPC Peering: Connect Two VPCs

Prove private connectivity with EC2, custom route tables, security groups, and a bastion host.

Student Name: Aayush Kumar

Batch: SEP batch_1 2026

AWS Region: ap-south-1 (Mumbai)

Completion Date: October 2026

1. Architecture Overview

The solution connects two isolated VPCs using an AWS VPC Peering connection with strict non-overlapping CIDRs and minimal least-privilege security group rules.

- **VPC A (10.10.0.0/16):** Contains a public subnet with an Internet Gateway and Bastion host (`10.10.1.10`), plus a private subnet with `ewb-app-a` (`10.10.2.10`).
- **VPC B (10.20.0.0/16):** Contains a private subnet with `ewb-app-b` (`10.20.1.10`) and no internet gateway.
- **Administration:** Managed from local laptop using SSH `ProxyJump` via `ewb-bastion`.

2. Resource Inventory & Address Plan

Resource Name	VPC / Scope	IPv4 CIDR / IP	Security Group
ewb-vpc-a	VPC A	10.10.0.0/16	-
ewb-a-public	VPC A	10.10.1.0/24	-
ewb-a-private	VPC A	10.10.2.0/24	-
ewb-vpc-b	VPC B	10.20.0.0/16	-

Resource Name	VPC / Scope	IPv4 CIDR / IP	Security Group
ewb-b-private	VPC B	10.20.1.0/24	-
ewb-bastion	ewb-a-public	10.10.1.10 + Public IPv4	ewb-sg-bastion
ewb-app-a	ewb-a-private	10.10.2.10 (No public IP)	ewb-sg-app-a
ewb-app-b	ewb-b-private	10.20.1.10 (No public IP)	ewb-sg-app-b
ewb-a-b-peer	Peering Connection	pcx-xxxxxxxx	-

3. Route Tables Configuration

Route Table	Associated Subnet	Destination	Target	Purpose
ewb-rt-a-public	ewb-a-public	10.10.0.0/16	local	Local VPC A communication
		0.0.0.0/0	ewb-igw-a	Bastion internet ingress/egress
		10.20.0.0/16	ewb-a-b-peer	Bastion SSH to App-B in VPC B
ewb-rt-a-private	ewb-a-private	10.10.0.0/16	local	Local VPC A communication
		10.20.0.0/16	ewb-a-b-peer	Forward path to VPC B
ewb-rt-b-private	ewb-b-private	10.20.0.0/16	local	Local VPC B communication
		10.10.0.0/16	ewb-a-b-peer	Return path to VPC A

4. Test Evidence & Terminal Verifications

Test 01: Baseline Failure (Pre-Peering Verification)

```
[ec2-user@ip-10-10-2-10 ~]$ hostname && hostname -I
ip-10-10-2-10.ap-south-1.compute.internal
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3072ms
```

✓ **Result:** Expected failure verified before peering routes were activated.

Test 02: Forward Path (App-A to App-B)

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=1.14 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.98 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.95 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.97 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
```

✓ **Result:** 4 replies, 0% packet loss.

Test 03: Reverse Path (App-B to App-A)

```
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=1.02 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.96 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.94 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.98 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
```

✓ **Result:** Bidirectional communication confirmed.

Test 04: Private SSH Access via ProxyJump

```
$ ssh ewb-app-b
Last login: Wed Oct 28 03:15:22 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$ who am i
ec2-user pts/0 2026-10-28 03:16 (10.10.1.10)
```

✓ **Result:** Shell established to App-B without exposing private keys to AWS.

Test 05: Controlled Failure & Recovery

```
# Step 1: Inbound ICMP rule removed from ewb-sg-app-b
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss

# Step 2: Inbound ICMP rule restored to ewb-sg-app-b
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss
```

5. Architecture Defense & Viva Answers

> **1. Why is the return route needed?**
>
> Network routing is directional. When App-A sends a packet to App-B, it reaches App-B through the peering link. When App-B replies, its routing engine consults its own route table. If no route exists for 10.10.0.0/16 pointing back to the peering connection, the packet cannot leave VPC B.

> **2. Why is the App-B SSH source the bastion private IP?**
>
> Using SSH ProxyJump, the bastion acts as an application-level proxy. The bastion terminates the first TCP leg and initiates a brand new TCP connection to App-B using its own private IP (10.10.1.10). Therefore, App-B's firewall sees the source as 10.10.1.10, not your laptop's public IP.

> **3. Why does Active peering not guarantee a successful ping?**
>
> Peering only establishes the network pipeline. Communication requires four aligned components: Active peering status, correct route table targets in both VPCs, explicit security group ingress rules, and permissive Network ACLs.

> **4. Can VPC B use VPC A's internet gateway over peering?**
>
> No. AWS VPC peering is strictly non-transitive. It does not support edge-to-edge routing, meaning traffic from one peered VPC cannot traverse another VPC to access an Internet Gateway, NAT Gateway, or VPN.

6. Cleanup Verification

All EC2 instances (`ewb-bastion`, `ewb-app-a`, `ewb-app-b`) were terminated. Peering connection (`ewb-a-b-peer`), custom route tables, Internet Gateway, and both VPCs were deleted to ensure 0 lingering charges.